

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Hard

Question

A right circular cone has a volume of $71,148\pi$ cubic centimeters and the area of its base is $5,929\pi$ square centimeters. What is the slant height, in centimeters, of this cone?

Answer

- A. 12
- B. 36
- C. 77
- D. 85

Correct Answer: D

Rationale

Choice D is correct. The volume, V , of a right circular cone is given by the formula $V = \frac{1}{3}\pi r^2 h$, where πr^2 is the area of the circular base of the cone and h is the height. It's given that this right circular cone has a volume of $71,148\pi$ cubic centimeters and the area of its base is $5,929\pi$ square centimeters. Substituting $71,148\pi$ for V and $5,929\pi$ for πr^2 in the formula $V = \frac{1}{3}\pi r^2 h$ yields $71,148\pi = (\frac{1}{3})(5,929\pi)(h)$. Dividing each side of this equation by $5,929\pi$ yields $12 = \frac{h}{3}$. Multiplying each side of this equation by 3 yields $36 = h$. Let s represent the slant height, in centimeters, of this cone. A right triangle is formed by the radius, r , height, h , and slant height, s , of this cone, where r and h are the legs of the triangle and s is the hypotenuse. Using the Pythagorean theorem, the equation $r^2 + h^2 = s^2$ represents this relationship. Because $5,929\pi$ is the area of the base and the area of the base is πr^2 , it follows that $5,929\pi = \pi r^2$. Dividing both sides of this equation by π yields $5,929 = r^2$. Substituting $5,929$ for r^2 and 36 for h in the equation $r^2 + h^2 = s^2$ yields $5,929 + 36^2 = s^2$, which is equivalent to $5,929 + 1,296 = s^2$, or $7,225 = s^2$. Taking the positive square root of both sides of this equation yields $85 = s$. Therefore, the slant height of the cone is 85 centimeters.

Choice A is incorrect. This is one-third of the height, in centimeters, not the slant height, in centimeters, of this cone.

Choice B is incorrect. This is the height, in centimeters, not the slant height, in centimeters, of this cone.

Choice C is incorrect. This is the radius, in centimeters, of the base, not the slant height, in centimeters, of this cone.